ESSENTIAL LECTURE 04

Photoshop Essentials

Prepare images for a clear and readable publication

Open • Crop • Resize • Save • Export



Four skills you will use today

After the lecture, you should be able to carry out a small image-editing task independently.

01 ORGANISE

Keep an untouched original and store each working file in a clearly labelled folder.

02 EDIT

Open an existing image, choose a crop and resize it without stretching.

03 CORRECT

Make a small brightness or contrast change only when the image needs it.

04 DELIVER • Save an editable Photoshop file and export a suitable JPEG or PNG.

Remember the A4 hand-out plan

Lecture 3 explained who the readers are and what the publication should communicate.

OUR RUNNING PROJECT

Tumaini Community Learning Centre

The one-page “Stay Safe Online” hand-out needs one appropriately prepared image to support the information, not to distract from it.

- The message is already planned.
- Now prepare an image to fit that message.
- Next lecture: combine it with text and shapes.

What is Photoshop used for?

In desktop publishing, image editing supports the quality and clarity of the final publication.

Photographs

Improve composition, crop unnecessary surroundings and make modest corrections to light or contrast.

Illustrations

Prepare raster illustrations to use in a poster, brochure or educational hand-out.

Working images

Export an appropriately sized copy while retaining the editable project and untouched source.

Photoshop mainly works with raster (pixel-based) images; it is not a replacement for every publishing application.

One workflow, different applications

The programs overlap, but each has a typical strength in a publishing project.

PHOTOSHOP

Prepare photographs, edit raster images and create pixel-based compositions.

ILLUSTRATOR

Draw scalable vector artwork such as an icon, logo or simple diagram.

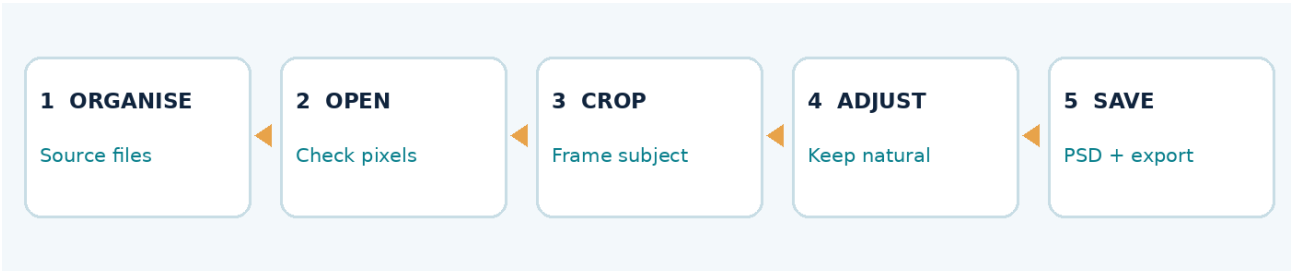
INDESIGN

Arrange text, image frames and graphics into the finished page layout.

In this lecture we use Photoshop for a small image task. Later lessons introduce the other applications at a basic level.

Five small steps—not fifty commands

Follow one repeatable route each time you prepare an image.



Think in sequence

Organise your files first, then inspect the image, crop only what is necessary, make modest adjustments and save both working and final output files.

A successful edit is usable, reproducible and easy to find.

Create a project folder before editing

One good habit can prevent much of the confusion beginners experience.

Tumaini_Handout

01_Originals

02_Working_Files

03_Exports

What each folder holds

Originals contain unchanged source images. Working_Files contains editable Photoshop projects. Exports contains JPEG or PNG files intended for the hand-out.

Never perform all of your changes on the only copy of the original image.

Use names that tell you what is inside

A clear filename matters when a practical project is submitted, revised or shared.

SOURCE IMAGE

Tumaini_Original_1600x1000.png

This filename identifies the starting artwork.

WORKING PROJECT

tumaini_image_edit.psd

This is the editable Photoshop project.

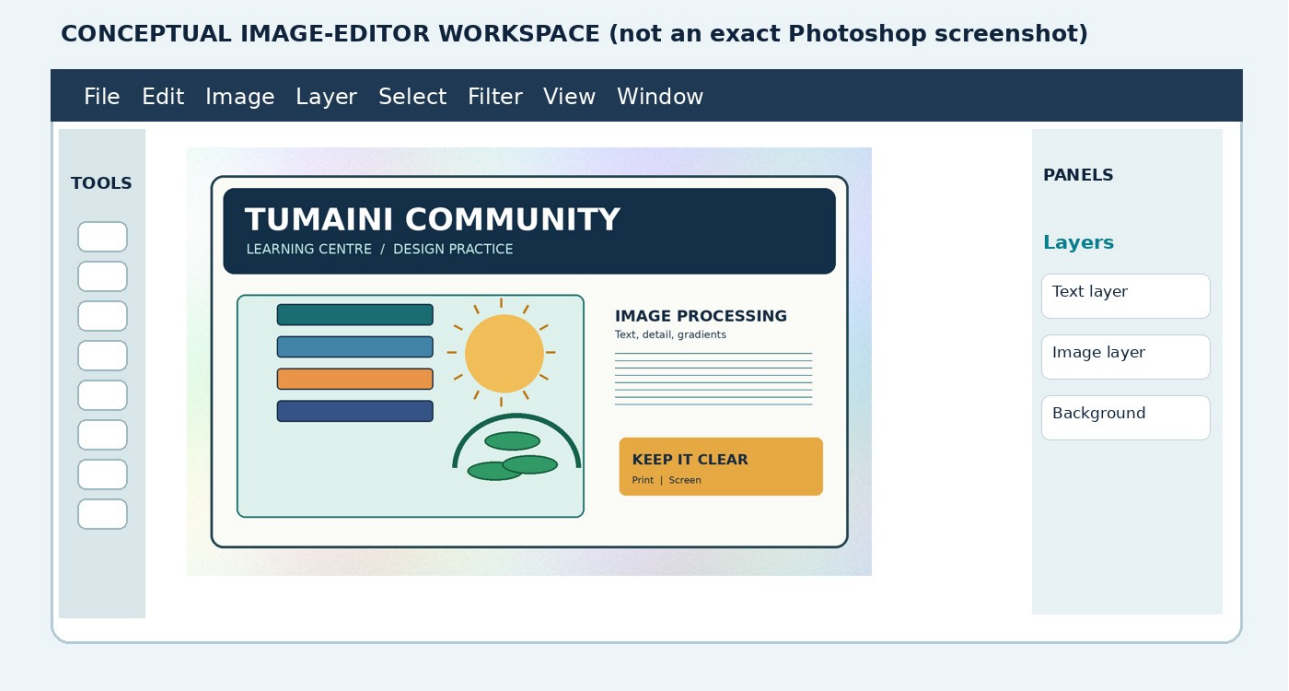
FINAL EXPORT

tumaini_image_web.jpg

This is the copy prepared for placing or sharing.

Recognising the Photoshop workspace

Menus, tools, panels and the image canvas each have different roles.



Read the labels

The Menu Bar contains commands, the Tools panel gives editing tools, the canvas displays the document, and panels show settings and image structure. The arrangement may vary by version.

Diagram is conceptual; not an exact Photoshop screenshot.

Four areas to locate before you begin

Finding these areas is more useful than memorising every button.

MENU BAR

File, Edit and Image contain commands for opening, adjusting and saving documents.

TOOLS PANEL

The Crop Tool changes image framing. Other tools are used for moving, selecting and editing.

DOCUMENT CANVAS

The canvas is the actual image or document area. The surrounding workspace is not part of the exported image.

Also locate the Layers panel: it shows the document components and becomes more important in Lecture 5.

File > Open or File > New?

These commands solve different problems.

File > Open

Choose Open when the image already exists. For this lecture, open the supplied Tumaini illustration or practice photograph and inspect its properties.

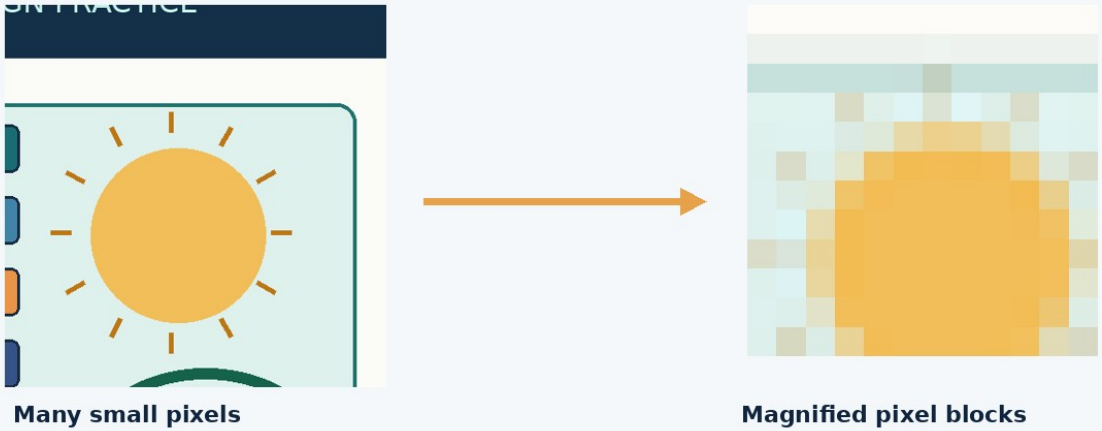
File > New

Choose New when you need a blank document, such as a new 1200 × 800 pixel graphic. Before starting, decide on width, height, units and intended use.

For the first practical, start with File > Open; do not complicate the task with a blank composition.

Raster images are made of pixels

Image detail is limited by the quality and dimensions of the source.



A pixel is a picture element

Pixels are tiny units that store colour values. Together, they form a raster image. Excessive enlargement can make blocks or softness visible because the missing original detail cannot simply be recovered.

Read width × height correctly

An image described as 1600 × 1000 pixels has two dimensions, not one overall size.

Worked example: 1600 × 1000 px

Width: 1600 pixels
Height: 1000 pixels
Total: 1600 × 1000 = 1,600,000 pixels

A pixel count describes the digital image, not its printed size.

Why it matters

A small image may look fine on a phone but become soft when enlarged in print. Check the intended placement size before choosing or resizing a file.

Image dimensions and print resolution

Pixels and pixels per inch (PPI) answer different questions.

Pixel dimensions

The numbers of pixels across and down the raster image. For example, 2400 pixels wide means the file contains 2400 columns of pixels.

Effective PPI

How many image pixels are allocated per printed inch. At 300 effective PPI, an image 2400 pixels wide covers 2400 ÷ 300 = 8 inches.

Changing a PPI label without resampling does not add new pixels.

Cropping changes what is visible

Cropping is about composition: keep the part of the image the reader needs.

WHY CROP?

Remove distracting surroundings, place the main subject more clearly and fit a chosen layout shape.

BE CAREFUL

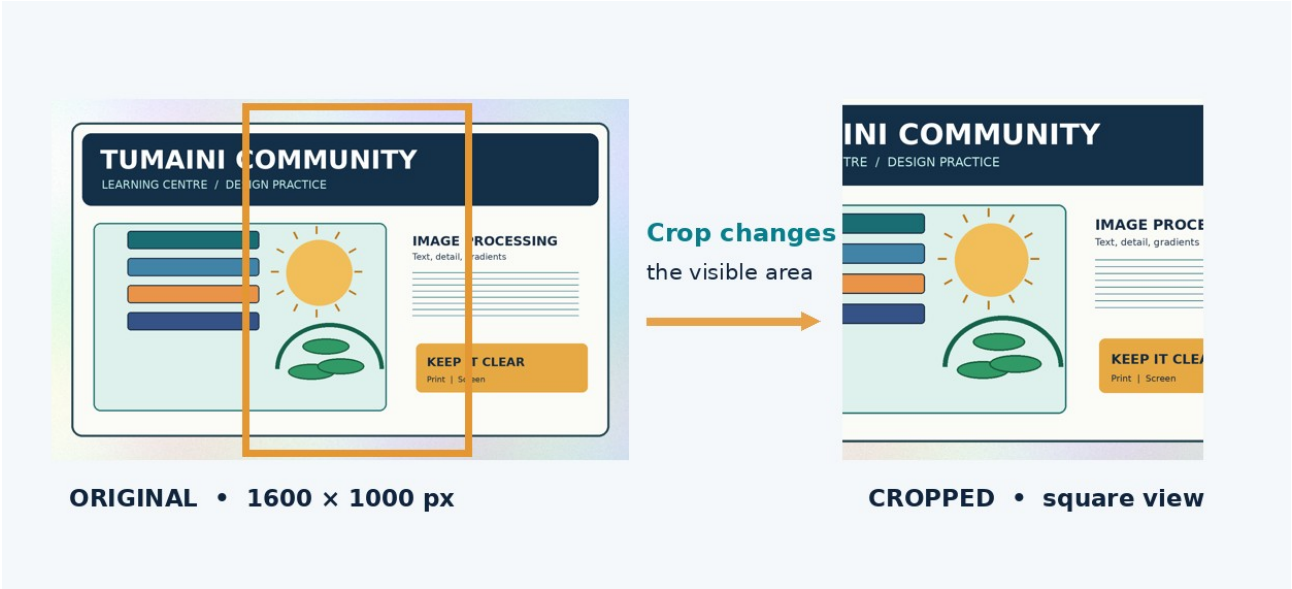
Do not remove an essential face, sign, instructional detail or visual clue by accident.

AFTER CROPPING

Check the new framing at the size readers are likely to see in the final publication.

Before and after: change the frame

A crop is not the same as simply shrinking the image.



See the difference

Cropping changes the visible scene. A square crop may help a design, but it can also remove important context. Always judge whether the image still communicates the intended idea.

A simple Crop Tool procedure

Follow the same sequence even if the exact buttons differ between Photoshop versions.

Do this

1. Open the supplied image.
2. Select the Crop Tool (commonly C).
3. Choose a suitable ratio or drag the handles.
4. Move the boundary and inspect what remains.
5. Confirm the crop and save a working copy.

Check before confirming

Is the main subject visible?
Is any important text cut off?
Does the framing fit the hand-out?
If “Delete Cropped Pixels” is available, consider leaving it off to preserve flexibility.

Aspect ratio describes shape—not pixel count

Different images can have the same aspect ratio but very different amounts of detail.

1 : 1 • SQUARE

800 × 800 px or 200 × 200 px. Often useful for a small icon or square image block.

4 : 5 • PORTRAIT

800 × 1000 px or 400 × 500 px. May suit a tall, narrow illustration.

16 : 9 • WIDE

1600 × 900 px or 800 × 450 px. Often used for landscape-shaped screen graphics.

Do not crop away the reason for the image

A technically neat crop is not useful if it destroys essential information.

Poor crop

A poster image shows a person pointing to a safety sign. Cropping away the sign leaves the gesture without a clear meaning.

Better crop

Keep the relevant subject and sign within the frame. Use spacing and alignment in the hand-out to make the most important information visible.

The reader’s need—not the decorative appearance—determines whether a crop is successful.

Resizing changes pixel dimensions

Resizing is different from cropping: it changes how many pixels the image uses.

Correct proportional resize

Original: 1600 × 1000 px
New width: 800 px
Scale factor: 800 ÷ 1600 = 0.5
New height: 1000 × 0.5 = 500 px

Prevent stretching

With width–height proportions linked, an object retains its natural shape. Changing width without adjusting height can distort faces, letters and symbols.

1600 × 1000 px resized to 800 × 500 px retains shape but has one quarter as many pixels.

Do not confuse Image Size with Canvas Size

One changes the image dimensions; the other changes the surrounding document area.

Image Size

Used to inspect or change an image's pixel dimensions. If resampling is enabled, Photoshop can add or remove pixels.

Canvas Size

Changes the area around existing content. For example, adding a border without scaling the original photograph is a canvas-size task.

For the basic practical: inspect Image Size; you do not need to master Canvas Size today.

What does “Resample” mean?

The number of pixels changes only when the software is told to recalculate the image.

Resample ON

Changing the pixel width or height alters the number of pixels. Reducing dimensions is common for smaller digital images. Enlarging does not recover genuine lost detail.

Resample OFF

If you change only the resolution value, the underlying pixel dimensions remain fixed. A different calculated print size is not the same as a new, sharper image.

Worked example: printing width

This simple calculation connects pixels to an approximate printed dimension.

2400 pixels ÷ 300 pixels per inch = 8 inches

A 2400-pixel-wide image covers about 8 inches when placed at 300 effective PPI.

Effective PPI is a planning guide; source detail, paper, printer and viewing distance still matter.

Make small changes to brightness and contrast

The goal is to make an image clear, not to create unnatural colour.

BRIGHTNESS

Changes the overall lightness of the image. Too much can wash out light areas and remove detail.

CONTRAST

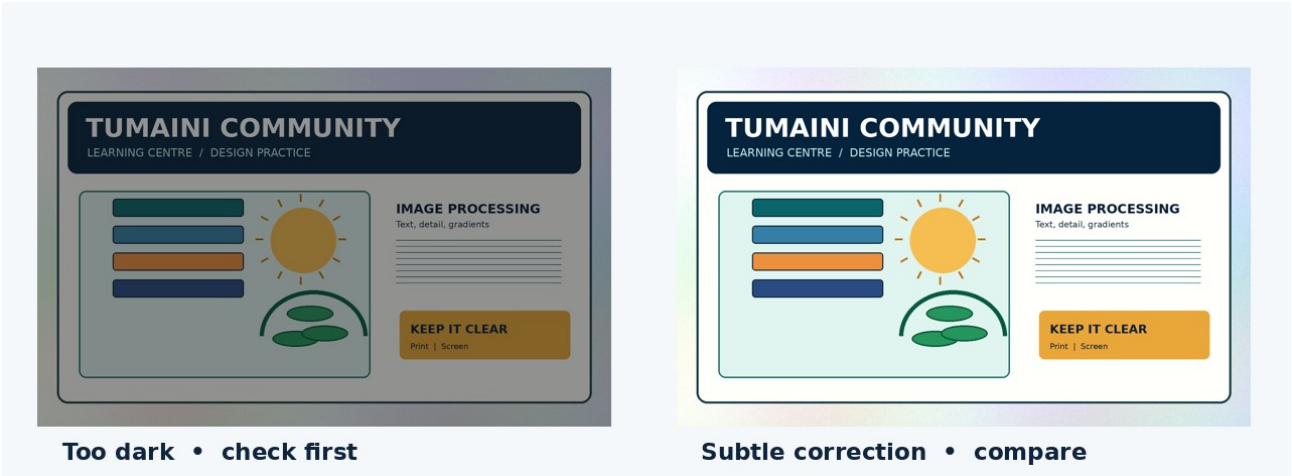
Changes the difference between light and dark regions. Too much may hide information in shadows.

COMPARE

Look at the original and the edited image together. Keep the correction only if readability improves.

A modest correction is usually enough

Compare the source against the edited image before deciding.



Too dark • check first **Subtle correction • compare**

Avoid heavy changes

If the image is dim, a modest correction may help. If it was already clear, leave it alone. An attractive-looking edit is not automatically better for communicating factual information.

A beginner-friendly adjustment procedure

Advanced non-destructive adjustment layers will be introduced in Lecture 5.

Try this in Photoshop

1. Duplicate or save a working copy.
2. Open Image > Adjustments > Brightness/Contrast.
3. Move the controls only a little.
4. Compare with the untouched source.
5. Undo the edit if details disappear.

Common warning signs

White areas lose their texture, dark areas become solid black, skin or objects look unnatural, or coloured text becomes difficult to read. Make a smaller change or return to the source.

Save the project before exporting

A Photoshop working file and an exported image serve different purposes.

Editable working file (.PSD)

Keeps the working document for future changes. If the project contains layers or editing information, a PSD can preserve them. Save under 02_Working_Files.

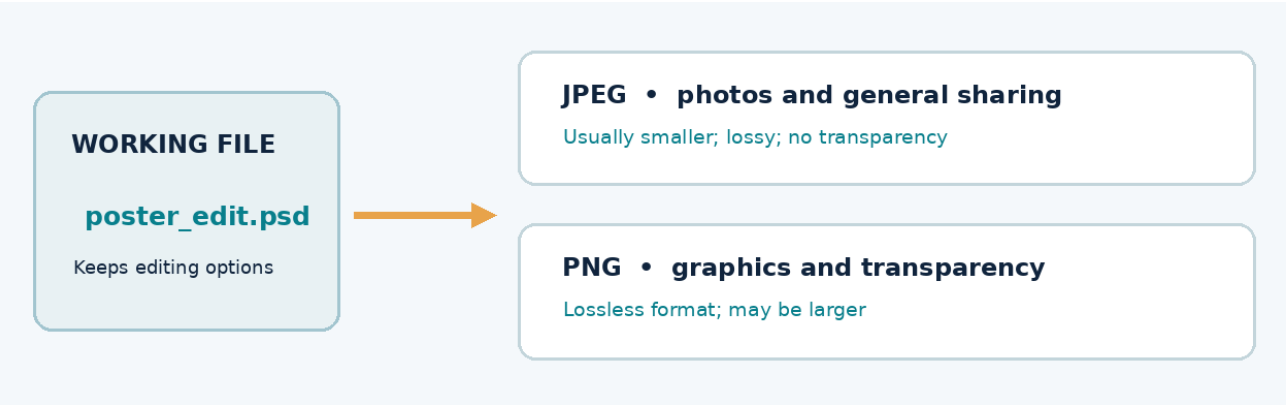
Exported file (.JPG or .PNG)

Intended for placing in a document, sharing or viewing. It may not retain the original editing structure. Save under 03_Exports.

Exporting is not a substitute for keeping an editable working project.

One project, two useful outputs

Preserve editability and prepare a sharing copy.



Choose the right purpose

PSD is the editable source for Photoshop work. JPEG is a common output for photographs and often produces smaller files. PNG can preserve transparency and sharp-edged graphic areas.

JPEG or PNG? Choose for a reason

A file format should suit the image and how it will be used.

JPEG

Often suitable for photographs and sharing. It uses lossy compression, and it does not support transparent image areas. Avoid repeatedly saving over the only image.

PNG

Useful for graphics, text-heavy images or transparent objects. It uses lossless compression. Some PNG files may be larger than comparable JPEG images.

Do not assume one format is always best. Test the image in the intended publication.

Where does each file belong?

Simple file organisation helps you recover, revise and submit your work.

01_Originals

Tumaini_Original_1600x1000.png

Unchanged source.

02_Working_Files

tumaini_image_edit.psd

Editable Photoshop work.

03_Exports

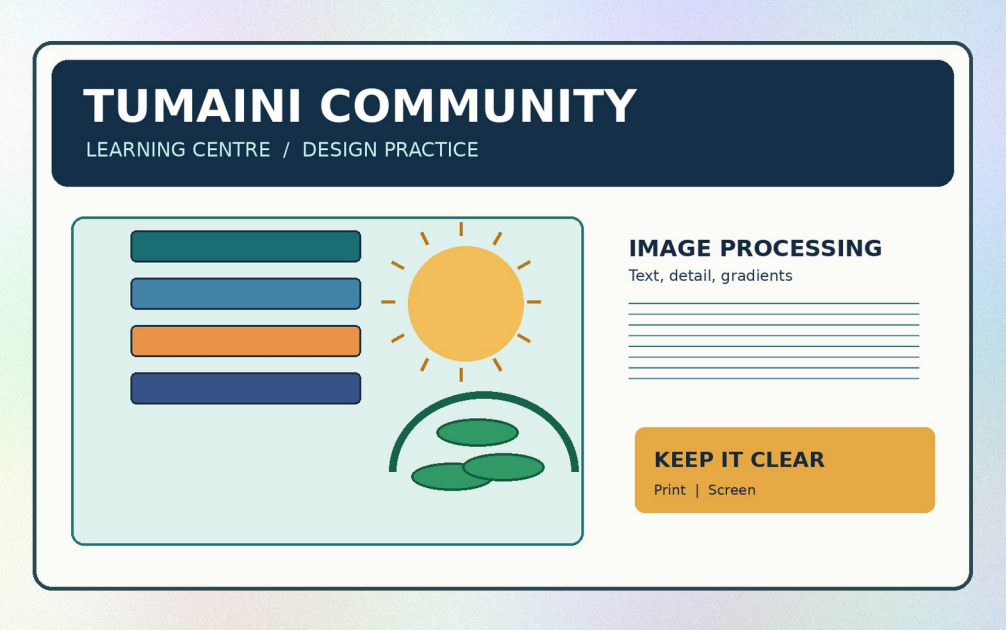
tumaini_image_web.jpg

Copy prepared for use.

Reopen the exported file before handing it in. Confirm the dimensions and appearance.

Step 1 — open and inspect the source

Start with a supplied image to reduce unnecessary setup problems.



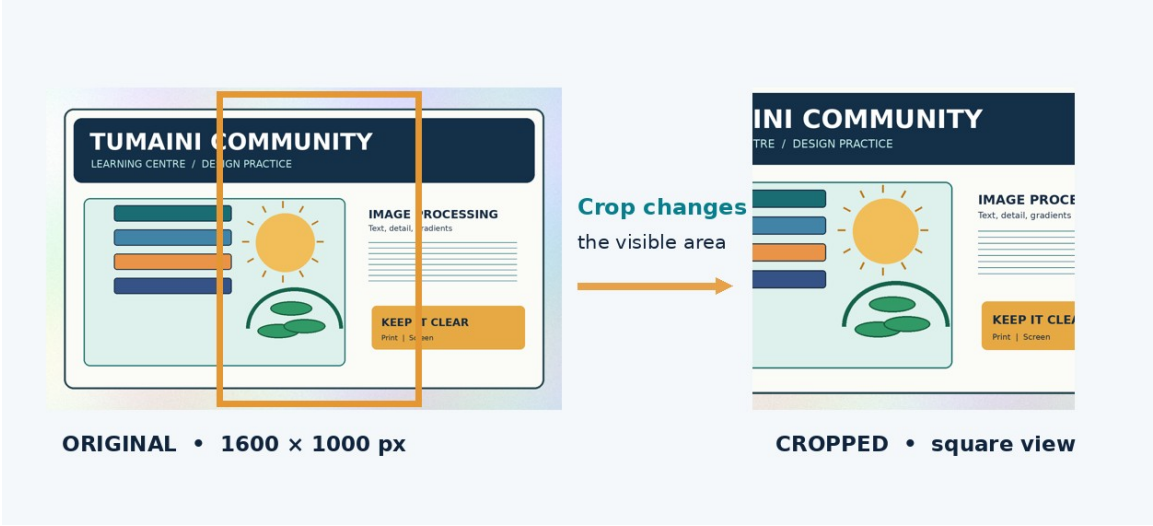
Follow the first actions

Make the three folders. Copy the original. Use File > Open. Check that the image is 1600 × 1000 pixels. Locate the Crop Tool and the Layers panel.

Begin with a source that opens correctly before attempting any editing.

Step 2 — choose and apply a crop

Your crop should help the reader see the subject clearly.



Do and check

Select the Crop Tool. Try a crop that keeps the main illustration. Check the edges carefully. Confirm only when the remaining image still conveys the intended message.

Step 3 — resize without stretching

The example exercise uses an 800-pixel-wide export.

FROM

1600 × 1000 px

→

TO

800 × 500 px

Keep the proportions linked. If the width is halved, the height must also be halved.

Step 4 — save and export separately

A small practical is complete only when its output can be reopened.

Save a working project

Use File > Save As or the available equivalent, and save a .PSD under 02_Working_Files. Keep the name descriptive.

Export a sharing copy

Use Export As, Save a Copy or the equivalent command to create a JPEG or PNG in 03_Exports. Reopen the file and check its size, crop and clarity.

Different Photoshop releases may label export commands differently.

Five mistakes beginners should avoid

Identify the likely cause before trying to fix the file.

STRETCHED IMAGE

Width changed without a matching height. Reopen Image Size and lock proportions.

LOST ORIGINAL

Only one file was edited repeatedly. Retrieve the source and create separate working/export folders.

BLURRY OUTPUT

Image enlarged beyond its detail. Use a stronger source or display/print it smaller.

Also check for over-adjustment and for forgetting to export a final JPEG/PNG.

Your short independent task

Prepare one image for the A4 “Stay Safe Online” hand-out.

Student checklist

- ☐ Original preserved
- ☐ Image opened and inspected
- ☐ Useful crop selected
- ☐ Dimensions changed proportionally
- ☐ Correction checked if used
- ☐ PSD saved and JPEG/PNG exported

Evidence to submit

One editable project; one exported image; a short completion record and two or three sentences explaining your crop and file format choice.

Check your understanding before the next lecture

Answer these without opening the software.

QUESTION 1

Explain the difference between cropping and resizing. Give one practical example of each.

QUESTION 2

A 1600 × 1000 image is reduced to 800 pixels wide. What is its proportional new height?

QUESTION 3

Why should a student submit both a PSD project and a JPEG or PNG export?

TAKEAWAY

What you should remember

Keep the image-editing process simple and purposeful.

Organise → Open → Crop → Resize → Check → Save → Export

An image is successful when it supports the publication, remains clear and can be edited again when needed.

NEXT: Use layers, shapes and text to produce a simple graphic composition.